

TABLE 9.—*Yields of winter wheat on plats following electrocultural treatment (positive charge), section B, Arlington Experiment Farm, in 1912*

Plat	Yields (pounds)			Ratio of treated to control		
	Shock	Grain	Straw	Shock	Grain	Straw
Treated.....	3,465	1,154	2,311	1.05	1.04	1.06
Control.....	3,300	1,114	2,186			

Experiments in 1913.—In the fall of 1912 the same plat in section B was again sown to wheat. The 7-foot network of the previous year was replaced by a permanent one 16 feet high, with cross wires 10 yards apart. The new network was erected over the northern half of the plat instead of the eastern half as in preceding years. The network was charged positively with the Lodge-Newman apparatus, and the treatment was given daily from 4 p. m. to 8 a. m.

The treated and control portions each had an area of three-fourths of an acre. At harvest the weights shown in Table 10 were recorded.

After the wheat was cut, cowpeas were sown on the B plat on July 29, 1913.

The static machine was connected to the network (16 feet high), giving about 40 to 50 kilovolts. The machine (positive charge) was run four hours a day from 3 to 7 p. m. for 32 days. On account of the lateness of the season, the cowpeas were cut for hay. After being stacked and cured, the crop was weighed in the field by means of a tripod and spring balance, showing the following yields: Treated portion, 1,807 pounds; control portion, 1,847 pounds; ratio of treated to control, 0.98.

TABLE 10.—*Yields of winter wheat on plats following electrocultural treatment (positive charge), section B, Arlington Experiment Farm, in 1913*

Plat	Yields (pounds)		Ratio of treated to control	
	Shock	Grain	Shock	Grain
Treated.....	3,254	808	1.04	1.03
Control.....	3,139	782		

Experiments in 1914.—Corn was planted in the B plat on May 24, 1914, and the network (16 feet high) was connected directly to one wire of a 6,600-volt 3-phase 25-cycle alternating-current power line running past the farm. The voltage was on continuously day and night for 110 days, when the corn was cut and the total weights recorded in the field. It was then shocked and given time to dry. Husking was done in the field on October 9, 1914, and the grain and fodder brought to a platform balance in the barn and weighed. The superintendent of the farm expressed the opinion that the treated plat had had some advantage over the check plat as regards soil-moisture conditions. The yields shown in Table 11 were recorded.